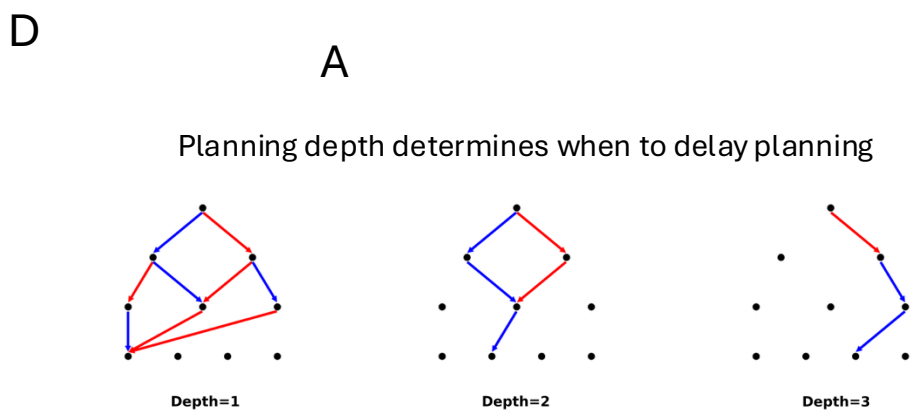
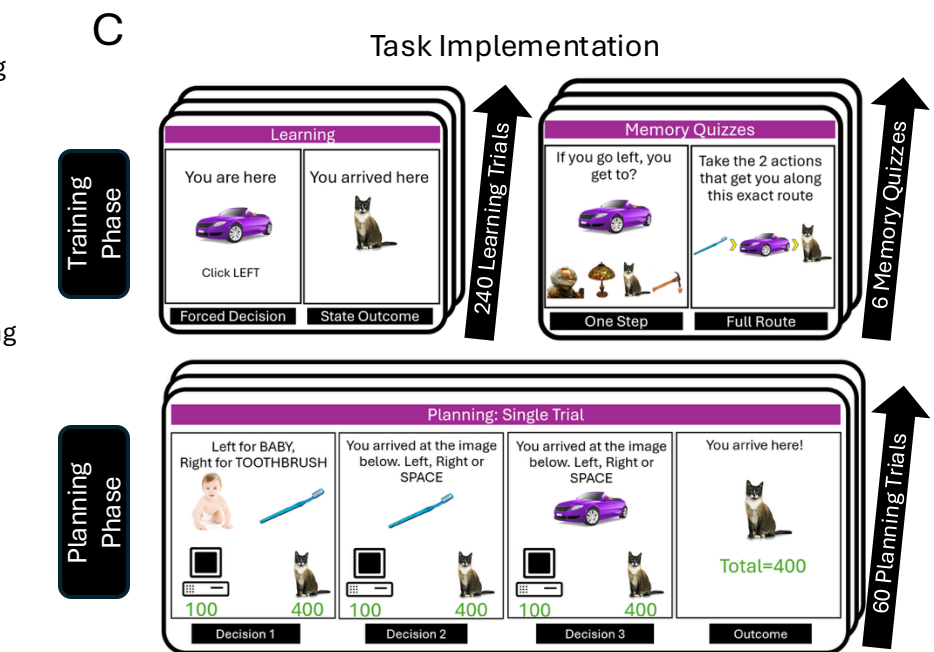
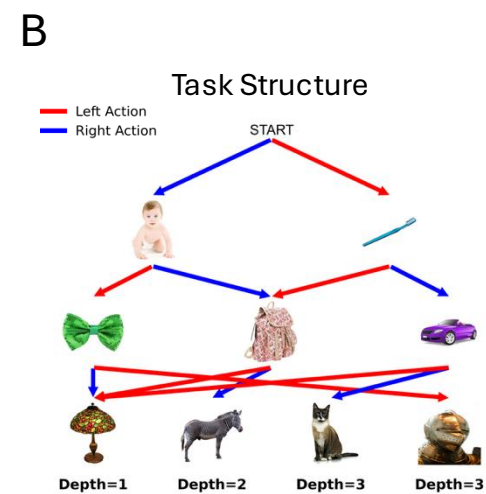
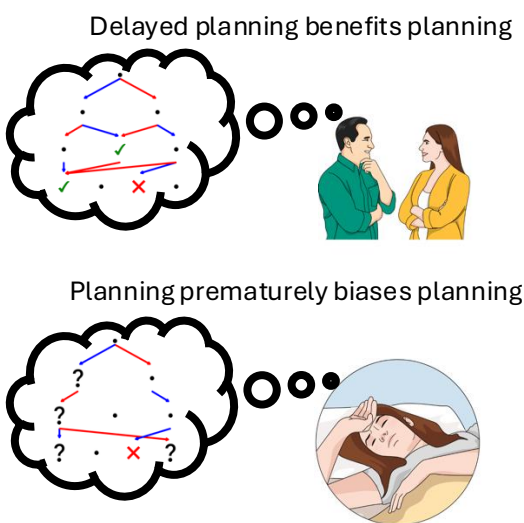
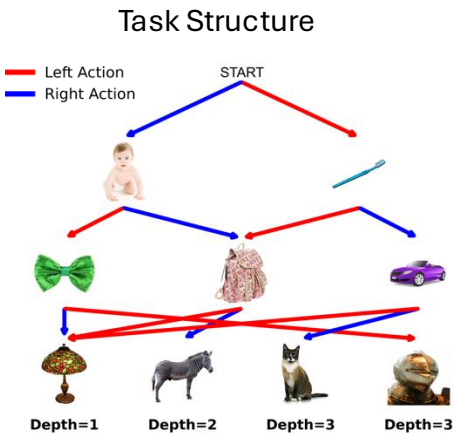
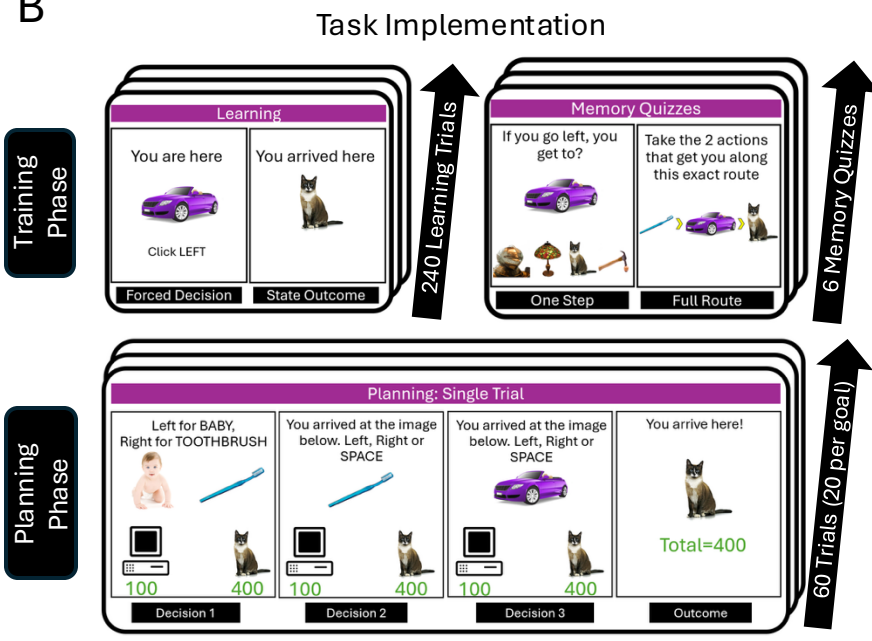




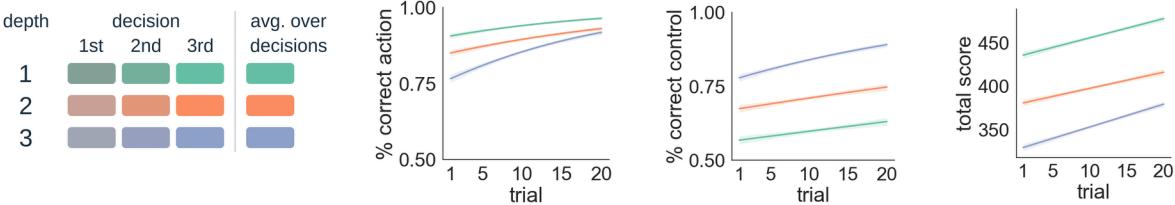
A



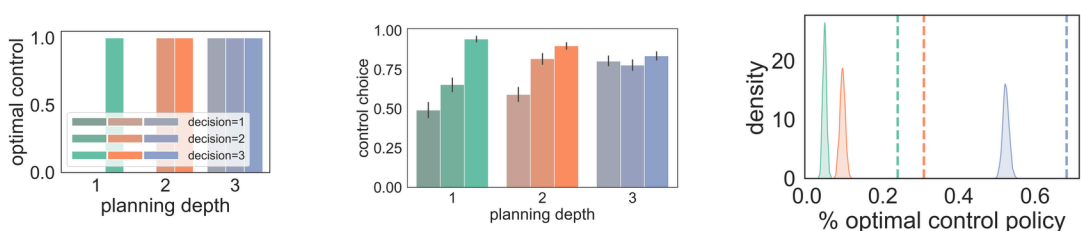
B



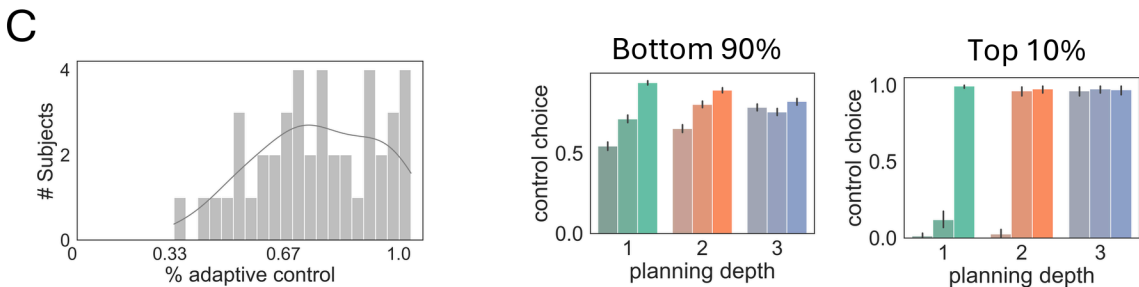
A Empirical task performance



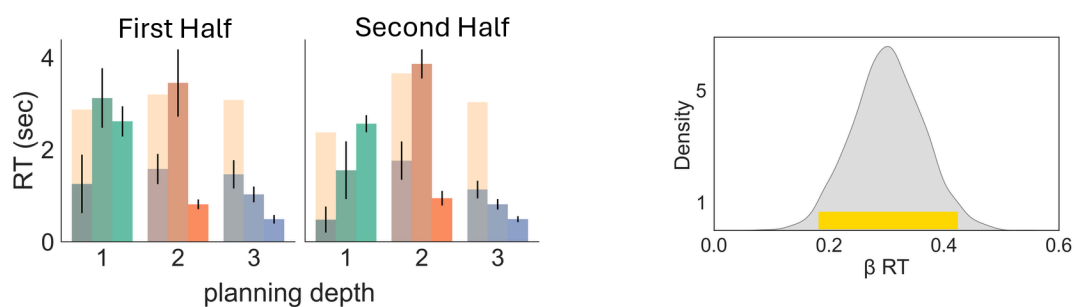
B Delayed planning in control choices



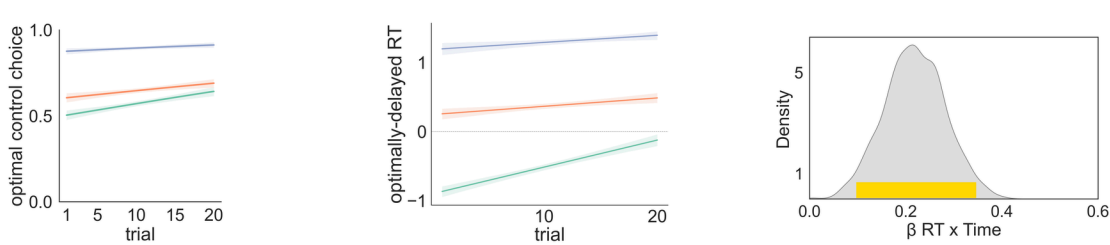
C Variability in control choices



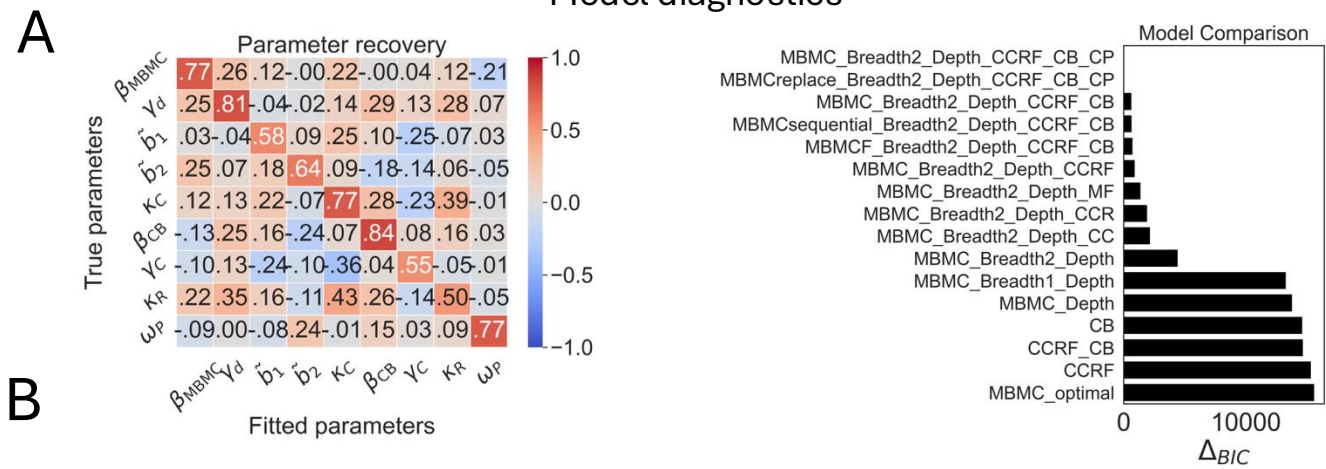
D Delayed planning in RT



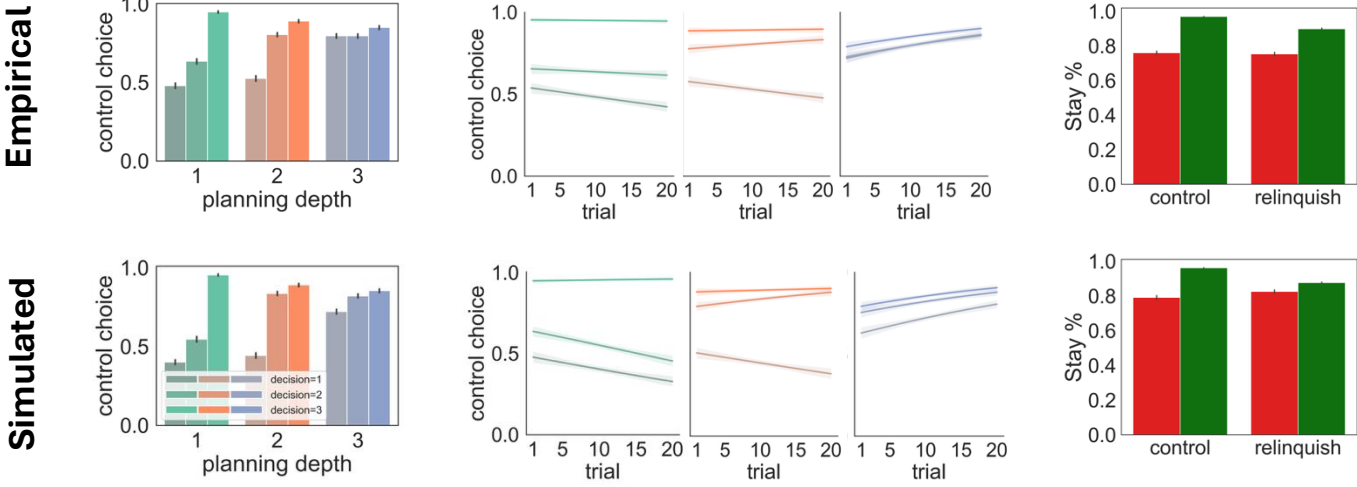
E Learning in control choice and RT



Model diagnostics



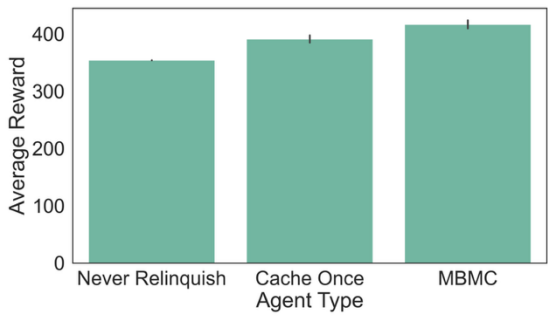
Computational model recapitulates behavioral trends



C

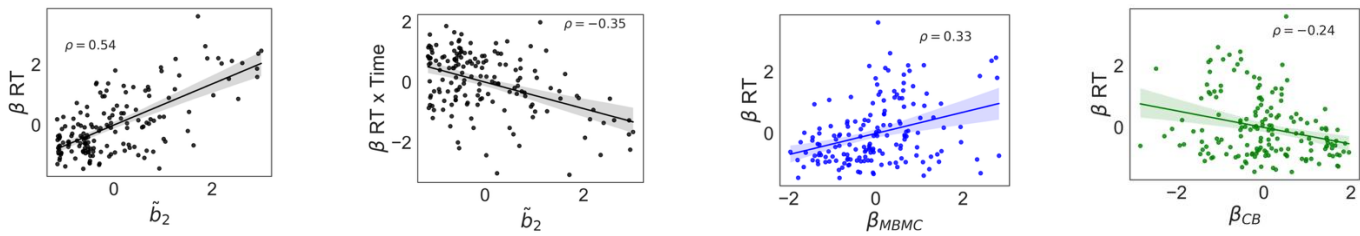
A

The benefit of delayed replanning in MBMC

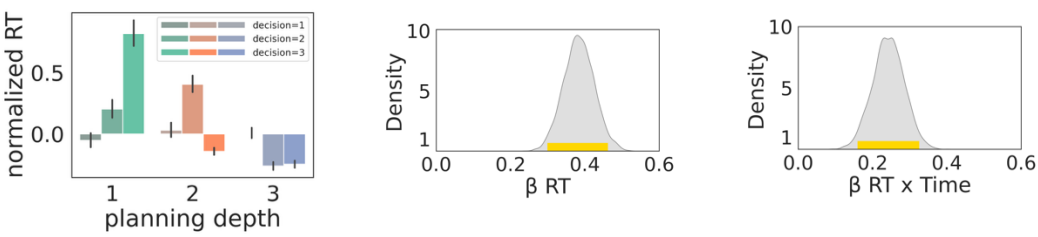


B

Computational model accounts for RT effects



Study 2 – replicating RT effects



Learning

You are here



Click LEFT

You arrived here



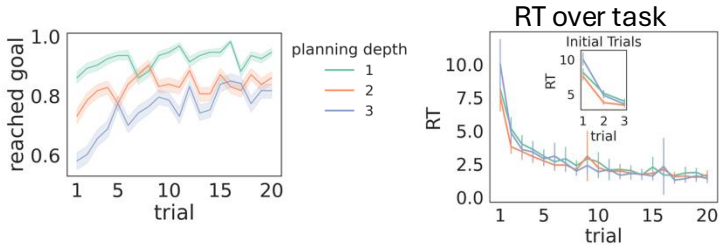
Forced Decision

State Outcome

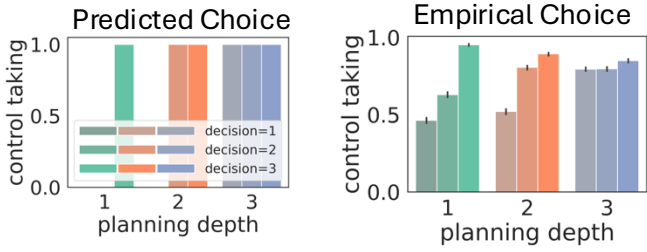
A

Task Performance

Goal attainment



A



C

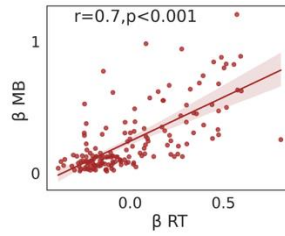
Computational Parameter

β_{MB} = Model-based weight

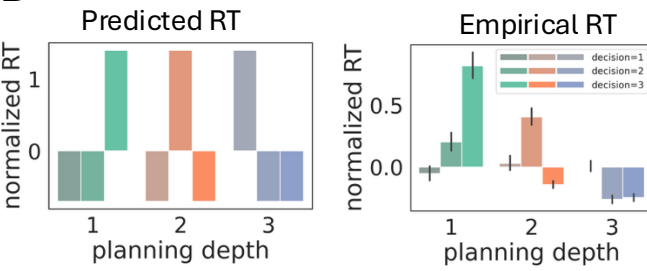
$$Q_{MB}(s, \tilde{a}) = \begin{cases} \text{Relinquish Control if} \\ Q_{plan}(s, a = 1)_t = Q_{plan}(s, a = 2)_t \\ \text{Take Control if} \\ Q_{plan}(s, a = 1)_t \neq Q_{plan}(s, a = 2)_t \end{cases}$$

D

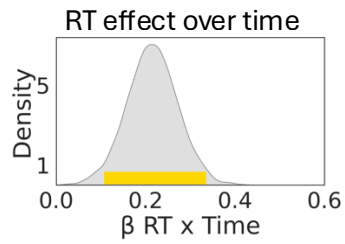
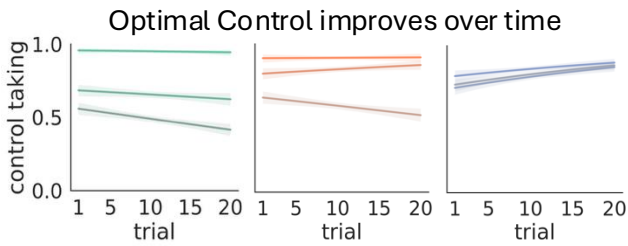
MBMC Parameter explains RT effect



B



Delayed planning is learned



Memory Quizzes

If you go LEFT here, you get to?



One Step

Take the two actions that get you along this exact route

Full Route

Memory Quizzes

If you go left, you get to?



One Step

Take the 2 actions that get you along this exact route



Full Route